Returns to Education in the Marriage Market: Access to Education, Age at Marriage and Assortative Matching in India

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Abstract

Using the staggered adoption of school construction across Indian districts for identification, I estimate whether increased school access had an effect on rates of educational attainment and marriage choices. I find that on average, an additional school built per 1,000 school-age children in a district is associated with 0.12 years more schooling for women in cohorts exposed to school construction. In districts exposed to school construction, age at marriage rises by 0.04 years, the age gap between husbands and wives falls by 0.03 years and the probability of child marriage drops by 0.3 percentage points. Marriage markets have become more segregated at the bottom of the education distribution, while assortative mating weakens at the high-school level over time. School construction increases educational sorting among couples by 0.03 as measured by the assortative matching index.

1 Introduction

In recent decades, India has witnessed a dramatic rise in educational attainment, particularly among women (Figures 1 and 2). This narrowing gender gap in schooling has occurred alongside persistently low levels of female labor force participation, raising an important puzzle. Conventional economic theory suggests that individuals invest in human capital in response to labor market returns. Yet, in the Indian context, where women's employment opportunities remain constrained by restrictive social norms and structural barriers (Mukherjee (2017), Afridi et al. (2023), Kleven, Landais and Leite-Mariante (2025)), the observed increase in female education seems to exceed what labor market incentives alone would predict. What, then, accounts for this apparent "over-investment" in women's education?

This paper investigates whether part of the answer lies in the marriage market. Marriage remains a near-universal institution in India, shaping both individual welfare and household economic outcomes. If education influences not only employment opportunities but also marital prospects, it may provide substantial non-labor returns. In particular, education may affect the probability of marriage, the timing and patterns of marital matching (Behrman et al. (1999), Peters (2002), Desai and Andrist (2010), Lafortune (2013), Royer and Geruso (2014), Bhaskar et al. (2015), Ashraf et al. (2020), Anderberg et al. (2022), Agarwal, Bahure and Javadekar (2023)), as well as the distribution of bargaining power and welfare within households (Choo and Siow (2006), Mourifié and Siow (2017), Doorley, Dupuy and Weber (2019)). Understanding these dynamics is crucial for evaluating the broader returns to education in settings where women's labor force participation is limited.

The mechanisms through which education may generate marriage market returns are multifaceted. Education can facilitate assortative matching, where men and women with similar educational attainment are more likely to marry, potentially due to shared preferences (consumption complementarities) or joint gains in household production and child rearing (production complementarities). Rising average schooling levels can also mechanically increase the likelihood of matches between educated partners, even without changes in sorting behavior. Moreover, assortative mating carries significant distributional implications: while it may improve coordination within households, it can also concentrate advantages across generations, exacerbating inequality and limiting mobility.

This study leverages large-scale variation in school construction across Indian districts and cohorts to identify the causal impact of expanded educational access on women's marriage outcomes. Specifically, I examine how exogenous increases in schooling opportunities influenced educational attainment, assortative mating patterns, age at marriage, and intra-household dynamics. By combining administrative data on school expansion with data from the Demographic and Health Survey, I assess both individual- and household-level consequences of education in the marriage market.

2 Background on school construction in India

School access has expanded dramatically in India since Independence (Figure 3). The Minimum Needs Program in the mid-1970s aimed to provide a primary school within 1.5 km of every village. According to census data, the

share of villages with primary schools increased from 53% in 1971 to 73% in 1991, middle schools from 8% to 19%, high schools from 4% to 8%. The District Primary Education Program (DPEP) in the mid-1990s increased funding for primary and upper-primary school building, hiring & training new teachers and decentralization of school administration. 160,000 new schools were built by 2000. This was followed by the Sarva Shiksha Abhiyan (SSA) in 2000s aimed at providing free and compulsory education for all children aged 6 to 14. By 2013, 98% of rural households had a primary school within 1 km and 92% had an upper primary school within 3 km. There has also been a huge increase in low-cost private schools across India in past few decades. Based on ASER data, 29% of children aged 6-14 in rural India attended fee-charging private schools in 2013.

3 Mechanisms

There are two key aspects to the characterization of returns to education in the marriage market: (i) marriage formation: trends in assortative mating (who marries whom with respect to education), and (ii) gains from marital surplus: payoffs to both genders within marriage conditional on educational attainment.

There are multiple mechanisms by which education could impact assortative matching in the marriage market. Schools and colleges act as matching platforms that reduce *search frictions* and enable matching in the marriage market (Kirkebøen, Leuven and Mogstad (2021)). This is less relevant in the Indian context where arranged marriages dominate. Individuals with similar

educational attainment may share similar preferences and derive greater utility from partnership due to *consumption complementarities* (Becker (1981), Choo and Siow (2006), Lundberg and Pollak (2012), Chiappori, Dias and Meghir (2018)). Couples with similar (or strategically complementary) education levels can achieve greater household efficiency through specialization in labor and better coordination in raising and educating children (shown empirically in papers by Behrman and Rosenzweig (2002), Azam (2015), Sunder (2019), Beauchamp, Calvi and Fulford (2021)). This may be referred to as *production complementarities* (Becker (1981), Calvo, Lindenlaub and Reynoso (2024)). There is also a *mechanical effect* wherein as education levels rise for all, individuals are more likely to match with other more educated individuals mechanically without a change in assortativity. There are also *general equilibrium effects* due to assortative mating in the form of spillovers on individuals not directly treated. Hence, it is crucial to use a measure of assortative mating that accounts for the distribution of traits in the population (Eika, Mogstad and Zafar (2019), Anderberg et al. (2022)).

A distinction must be made between educational homogamy vs assortative mating. Changes in educational homogamy (fraction of couples with the same level of education) may be due to secular changes in educational attainment or due to changes in educational assortative mating. To deal with this issue, I use a definition of assortative mating that accounts for shifts in the marginal distribution of education – men and women with same level of education marrying more or less frequently than at random in terms of education following Eika, Mogstad and Zafar (2019).

4 Contribution

There is an extensive literature on returns to education in the labor market. However, the literature on marriage market outcomes is fairly nascent. The latter is particularly consequential in the Indian context, where female labor force participation remains among the lowest in the world, raising the question of whether returns to education for women are channeled through the marriage market rather than direct participation in the labor market.

With respect to estimating labor market returns in a developing country context, Duflo (2001)'s paper is significant. It evaluates the impact of a massive primary school construction program that took place between 1973 and 1978 in Indonesia using a difference-in-differences design that exploits variation across regions and birth cohorts. Each additional school built per 1,000 children led to an increase in average educational attainment of 0.12 to 0.19 years and wage increases of 1.5 to 2.7 percent. In my paper, I follow the methodology set out in Duflo (2001) using a long panel. My results on educational attainment are in line with Duflo (2001) but I also study the effect on marriage market outcomes.

In the Indian context, the closest papers to this one are by Khanna (2023) and Agarwal, Bahure and Javadekar (2023) which study the effect of primary school construction on men's labor market outcomes and women's marriage outcomes respectively. Both studies use a regression discontinuity design that exploits district-level cutoffs for increased funding for school construction during the 1990s. The regression discontinuity estimand provided by Khanna (2023) and Agarwal, Bahure and Javadekar (2023) is a local treatment effect for districts with female literacy around the cutoff whereas the

difference-in-differences specification I use exploits variation across all districts and cohorts. Khanna (2023) and Agarwal, Bahure and Javadekar (2023) also focus specifically on the District Primary Education Program (DPEP) expansion in the 1990s whereas I use school construction data over a longer time frame. My dataset also includes construction of all schools and not only primary schools. In contrast to Khanna (2023) who focuses on labor market outcomes, I study the effect of school construction on the marriage market. Khanna (2023) finds that the DPEP led to 0.3 more years of schooling for men but only 0.09 more years for women. My study finds school construction led to a higher estimate of 0.12 more years of schooling for women.

Agarwal, Bahure and Javadekar (2023) find that program exposure raised educational attainment by 0.4 to 1 year, but unexpectedly reduced age at marriage by 0.75–1.55 years. Despite earlier marriage, women exposed to DPEP experienced lower fertility, reported less domestic violence, and matched with wealthier spouses. My study complements Agarwal, Bahure and Javadekar (2023) in studying the effect on marriage outcomes but with different empirical results – notably Agarwal, Bahure and Javadekar (2023) find a counter-intuitive lowering of the age at marriage whereas I find a rise in the age at marriage. The effects on educational attainment found by Agarwal, Bahure and Javadekar (2023) are also much larger than the effects found in other studies including this paper.

I also build an assortative mating index for India documenting an increase in segregation of individuals with no education into their own closed marriage market, during a time of expanding school construction. While I follow

the methodology of Eika, Mogstad and Zafar (2019), this measure has not previously been constructed for India. Earlier studies use a reduced form method correlating education levels of husbands and wives, which ignores changes in the marginal distribution. In addition, I also show that school construction increases educational sorting among couples by 0.03 as measured by the assortative matching index. While prior studies such as Anderberg et al. (2022) and Royer and Geruso (2014) have studied the effect of schooling on matching (both use the Raising of School-Leaving Age in the UK in 1972), I am able to improve upon these studies by exploiting both the spatial and temporal dimension of school construction programs instead of only variation across cohorts at the time of the policy change.

5 Data

Data on school construction and locations is available from Department of Education's UDISE database. UDISE is a comprehensive database of all 1.5 million public and private schools in India and includes detailed data on many school-level variables. However, I only use data on school location and year of construction to construct my treatment variable. The UDISE data was collected in 2018 through school surveys but my dataset includes all schools built since 1960. One concern is that this dataset misses any schools that had shut down before the time of survey.

I use education and marriage data from India's Demographic and Health Survey (DHS), a large-scale nationally representative survey of women aged 15-49 years. Since district identifiers are only available from 2015 onwards, I

use data from the 2019 survey round.

Population data was drawn from the decadal variation in population tables published by the Census of India and then linearly interpolated across years within decades. These tables account for boundary changes while estimating population within a district.

A key challenge in conducting multiple-year analyses at the district level in India is reconciling the many changes in state and district boundaries that have occurred since the first comprehensive census of independent India in 1961. A key contribution of this paper is construction of this long-run district-level panel.

6 Empirical Strategy

School construction in India is plausibly exogenous since it was governed by centralized population-and-distance norms under successive Five-Year Plans and later Sarva Shiksha Abhiyans, with funding tied to predetermined district infrastructure targets rather than contemporaneous local education demand or marriage-market conditions. Once built, schools represent a long-lived, fixed infrastructure investment, making reverse causality from cohort-specific marriage outcomes implausible. Also, as exposure is determined by district of childhood residence and the timing of school construction during ages 6–15, treatment variation is fully predetermined with respect to adulthood marriage outcomes. Migration at school ages is rare, and outcome realizations occur many years after treatment. While women do migrate at the time of marriage, marriage outside the district is rare according to Census

data.

Specifically, I estimate the impact of school construction on the following outcomes for women – educational attainment (years of schooling, educational qualifications), not married by age x (extensive margin), and age at marriage. I also estimate the effect for couples on the age gap (across-cohort sorting) and the educational assortative mating index (intensive margin).

Following Duflo (2001), I use a two-way fixed effects estimation specified as:

$$\mathrm{Y_{idt}} = \beta(I_{\mathrm{d}}{*}C_{\mathrm{i}}) + \gamma_{\mathrm{t}} + \delta_{\mathrm{d}} + \varepsilon_{\mathrm{idt}}$$

$Y_{\mathbf{idt}}$ is outcome of individual i in district d born in year t, $\delta_{\mathbf{d}}$ is district fixed effect, $\gamma_{\mathbf{t}}$ is cohort fixed effect, $I_{\mathbf{d}}$ is treatment intensity (number of schools per 1000 school-aged children) in district d, $C_{\mathbf{i}}$ dummy equals 1 for exposed cohort, 0 for unexposed cohort. The exposure window for a cohort is considered to be between 6 to 15 years – this means that a woman's degree of treatment exposure or dosage is determined by the number of schools in the district she resided in when she was between 6 to 15 years old.

I use a continuous treatment measure since the policy generated large heterogeneity in construction intensity across district–cohorts, so a binary exposure dummy would mechanically understate the true effect by discarding economically meaningful variation. Following Duflo (2001), I also present results with a binary treatment where 1 indicates more schools constructed than the (cohort-specific) mean and 0 fewer schools than the mean. The results are qualitatively similar.

Changes in educational homogamy (fraction of couples with same level of education) may be due to secular changes in educational attainment or due

to changes in educational assortative mating. Instead, my empirical strategy follows Eika, Mogstad and Zafar (2019) and uses a definition of assortative mating that accounts for shifts in the marginal distribution of education – men and women with same level of education marrying more or less frequently than at random in terms of education. This is a structural assortative mating index which is free of marginal education distribution bias. Marital sorting between a woman with education level e_f & a man with education level e_m is defined as the observed probability that a woman with education level e_f is married to a man with education level e_m relative to the probability under random matching with respect to education.

$$s(e_f, e_m) = \frac{Pr(E_f = e_f, E_m = e_m)}{Pr(E_f = e_f)Pr(E_m = e_m)} \tag{1}$$

The sorting index equals 1 under random matching and an index >1 indicates positive assortative matching. The higher the index, the stronger the extent of assortative matching.

7 Results

Table 1 shows the effect of school construction on woman's own educational outcomes and Table 2 on marriage outcomes. The treatment intensity measure in Panel A is the number of cumulative schools built per school-aged population in a woman's district by a given year and district. The initial level of schools is absorbed by district fixed effects. In all regressions, I include district and cohort fixed effects and cluster standard errors at the district level.

I find significant effects on women's own educational attainment. Table 1 Panel A shows that an additional school built per 1,000 school-age children in a district is associated with 0.12 years more schooling for women in cohorts exposed to school construction. The mean years of education for the estimation sample was 6.9 years. Increased school construction during a woman's schooling years also significantly raises the likelihood of her completing primary schooling by 1.9 percentage points. However, there is no significant change in secondary schooling rates.

In Table 1 Panel B, I also show results using a binary treatment where 1 indicates schools higher than cohort-specific mean and 0 indicates lower than cohort-specific mean. The results are qualitatively similar for educational outcomes – moving to a district with higher number of schools than the cohort-specific average from one with lower than average schools results in 0.4 additional years of schooling.

Table 2 shows the effect of school construction on a woman's age at marriage, the age gap between husbands and wives, and the probability for a woman not being married by age 18. I find statistically significant effects on all outcomes at the 5% level. School construction is associated with a rise in the age at marriage by 0.04 years whereas the mean age at marriage for the estimation sample was 19 years. The age gap between husbands and wives falls by 0.03 years (mean age gap is 4.8 years). In order to calculate the age gap between husband and wife, I supplement the data on husband's age from the women's survey (which has many missing values) with data from the household roster, however the latter is only available for co-residing spouses. I also cross-validate the spousal age distribution in the household

roster data with the subset of observations with non-missing values in the women's survey and find that the correlation is close to 1. The probability of child marriage falls by 0.3 percentage points (64% of sample is unmarried by age 18). Panel B shows the results for the binary treatment – again, the results are qualitatively similar. Moving to a district with higher number of schools than the cohort-specific average from one with lower than average schools results in age at marriage being 0.28 years higher, age gap being 0.1 year lower and the probability of being unmarried by age 18 becomes 2.3 percentage points higher.

Table 3 shows results from a regression of husband's educational outcomes where the treatment is a woman's exposure to school construction during childhood. While there is a positive effect on husband's educational attainment, the effects are not significant except for secondary schooling completion in the specification with binary treatment. This pattern is consistent with marriage-market selection. As women's educational attainment rises, particularly at key thresholds, marriage markets re-sort such that more educated women match with more educated men. The fact that the effect appears at the secondary margin — where male educational attainment is more heterogeneous — supports an assortative matching interpretation. However, it should be noted that husbands' and wives' exposure to education to school construction may be highly correlated since they live in the same district (data on women's childhood residence is not available) and age gaps between couples have been narrowing over time resulting in treatment exposure occurring at the same time for men and women in more recent years. As a result, the estimates in Table 3 should not be interpreted as causal effects

of women's education on men's schooling. Instead, they reflect equilibrium outcomes in local marriage markets where both men and women may have been exposed to educational expansion. Consistent with this interpretation, I focus on market-level measures of assortative mating in Table 4, which explicitly account for joint exposure and capture changes in sorting behavior net of changes in marginal educational attainment.

Figure 5 shows trends in assortative mating over marital cohorts. Low-education individuals in India are increasingly segmented into internally homogeneous marriages, far beyond what education expansion alone would generate mechanically. For individuals with no education, the index starts around 1.2–1.4 in early 1980s, rises almost monotonically and reaches 5–5.7 by late 2010s. This may imply that non-educated individuals increasingly form a closed marriage market as education expands for others. This is structurally identical to what Eika, Mogstad and Zafar (2019) finds for the U.S. low-educated after 1980, but the magnitude here is much larger. Individuals with primary education also assort positively but are less sharply stratified than the no education group. The upper primary group has positive but weaker and fairly stable assortative mating. For high school group, there has been a decline in assortative mating in recent years. As education expands, high school may be becoming the new "mass" category, not an elite one, so homogamy weakens. Again, this mirrors the decline in high-skill assortative mating that Eika, Mogstad and Zafar (2019) document for U.S. college graduates after 1980. Overall, India's marriage market is becoming dramatically more segregated at the bottom of the education distribution, while assortative mating weakens at the high-school level. This may imply rising

intergenerational persistence of low human capital, even as overall schooling expands.

I also estimate the effect of school construction on the educational assortative mating index. Table 4 reports two-way fixed-effects estimates of the effect of district-level school construction on the assortative mating index by marriage cohort. The coefficient on school exposure is positive and statistically significant at the 1 percent level. Quantitatively, an increase of one unit in school construction exposure raises the assortative mating index by approximately 0.03 relative to a baseline level of 2.1. Note that the baseline level of the index indicates that India's marriage market is highly positively assortative even absent new schools. This implies that education expansion through school construction leads to a strengthening of educational sorting in marriage markets. The result suggests that schooling policies alter the general-equilibrium structure of matching, potentially amplifying human-capital stratification across households.

8 Conclusion

I find that school construction raises women's own educational attainment and age at marriage. My results on educational attainment are slightly larger than Khanna (2023) who finds an increase of 0.09 years using National Sample Survey data but smaller than estimates in Agarwal, Bahure and Javadekar (2023) who find an increase of 0.32 or 1 year (depending on sharp or fuzzy RDD respectively) using DHS data similar to this study. It should be noted that comparing my estimates with theirs is not strictly appropriate since

these represent different estimands. The regression discontinuity estimates in Khanna (2023) and Agarwal, Bahure and Javadekar (2023) identify a local average treatment effect of DPEP eligibility for districts at the female literacy cutoff. In contrast, my difference-in-differences design exploits continuous variation in school construction intensity across districts and cohorts over a much longer horizon, identifying an average marginal effect of additional school availability. Since DPEP represented a large, discrete expansion targeted at low female literacy districts with high marginal returns to schooling, RDD estimates are expected to be larger than the average marginal effects estimated here. Also, increased school funding under DPEP may have been used for teacher hiring or infrastructure upgrades in existing schools in addition to new school construction whereas my estimates reflect the effect of constructing an additional school alone.

With respect to marriage outcomes, Agarwal, Bahure and Javadekar (2023) counter-intuitively find a fall in age at marriage by 0.75-1.55 years whereas I find an increase of 0.04 years. The observed increase in age of marriage in my study could reflect an enrollment or incapacitation effect of schooling — girls remain unmarried simply because they are still attending school which is less likely to matter at the primary school level (except for rare instances of child marriage). It could also be the case that a change in marriage or partner preferences occurs at a later stage of schooling.

My results on assortative mating trends mirror those found in the US data by Eika, Mogstad and Zafar (2019) re-iterating a pattern of increased stratification in marriage markets at the bottom of the education distribution. I also find that school construction is not flattening the marriage market. It

is instead increasing educational sorting among couples. In future work, I plan to look at the effect on intergenerational persistence of education using IHDS panel data to verify if these these trends in educational stratification in the marriage market persist through children's education.

To summarize, I find that school construction increases women's educational attainment and generates systematic changes in marriage timing and matching patterns – women marry later, are less likely to enter child marriage, and match with husbands closer to their own age. These shifts are consistent with education raising women's value in the marriage market, thereby improving their outside options and allowing families to delay marriage in search of higher-quality matches. The observed changes in assortative mating further indicate that schooling expansions reshape the equilibrium structure of marriage markets by altering the distribution of educated women, strengthening educational sorting at the bottom of the distribution while weakening assortative matching at higher education levels where schooling becomes less scarce. Taken together, these findings imply that increased access to schooling generates meaningful marital returns to education in equilibrium, even in settings where labor-market returns to female education are limited.

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9 Figures and tables

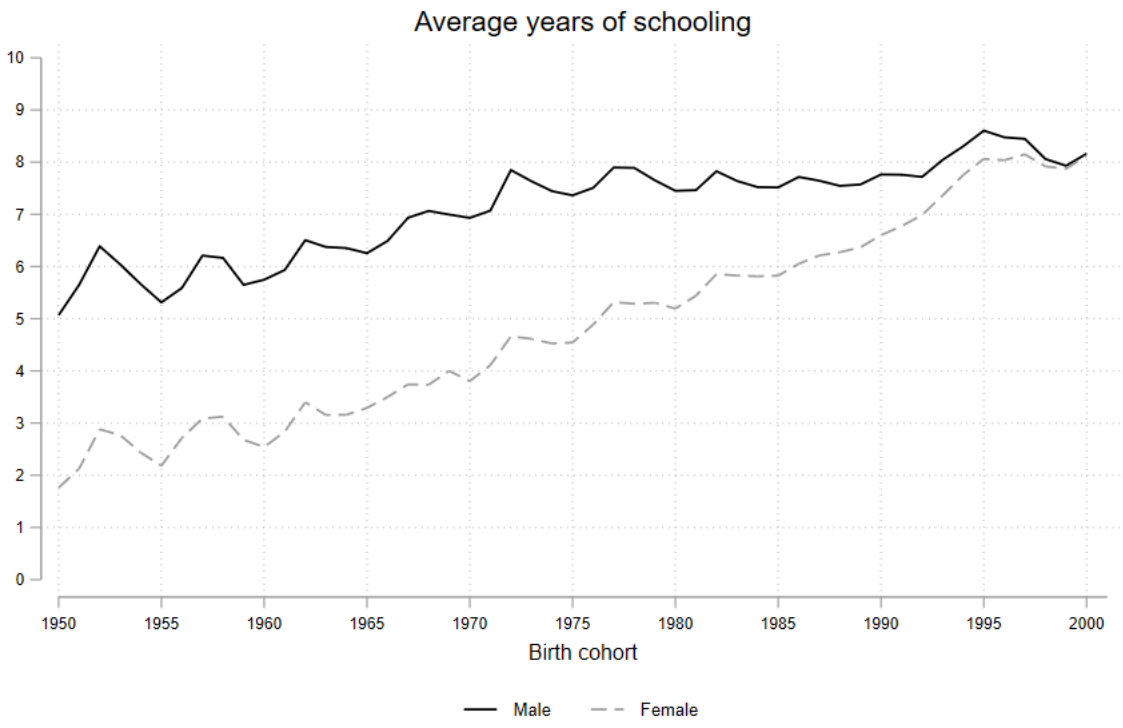

Figure 1: Average years of schooling by birth cohort

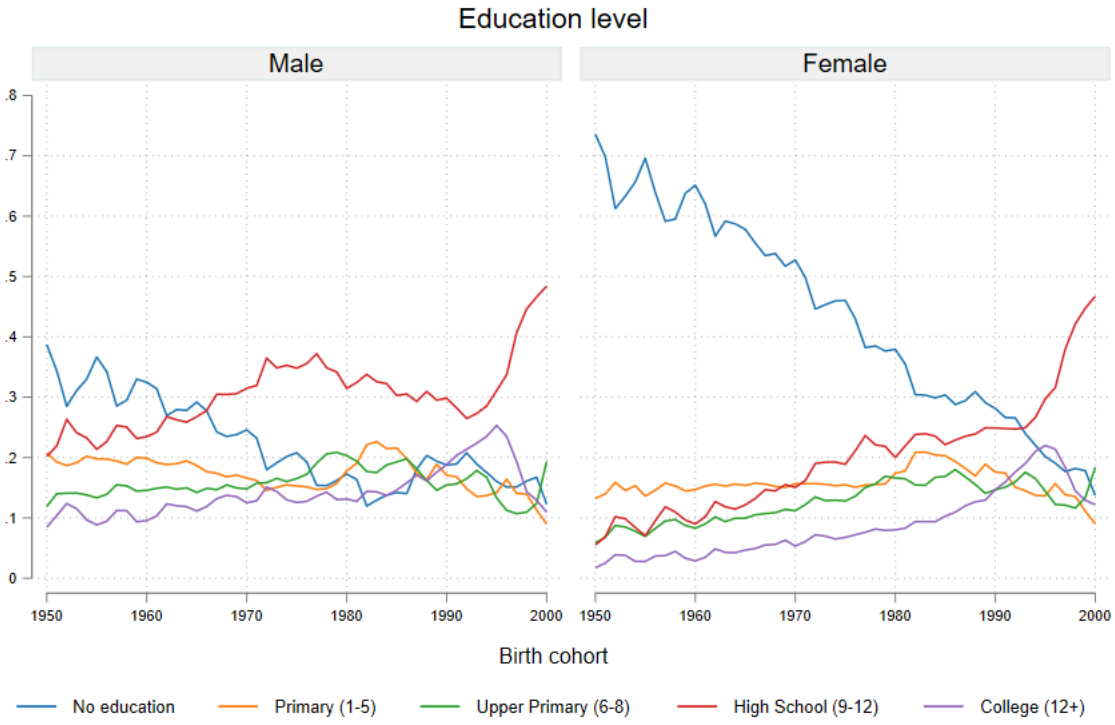

Figure 2: Education level (DHS 2019)

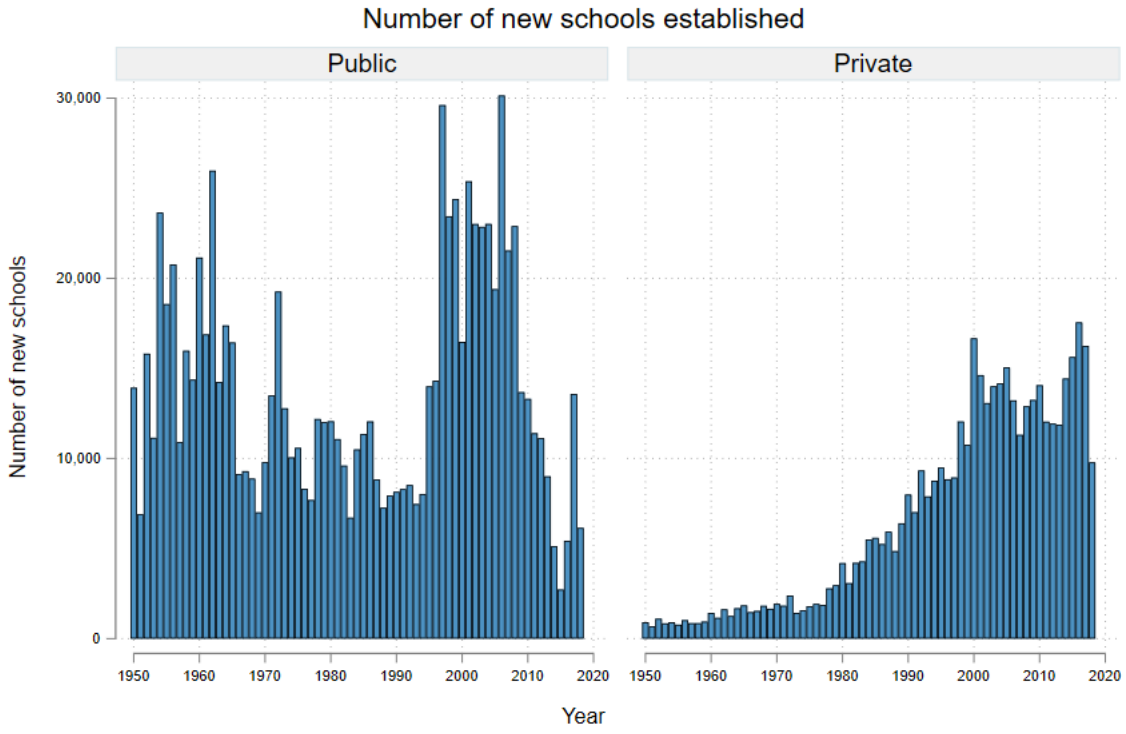

Figure 3: School construction over time (UDISE 2018)

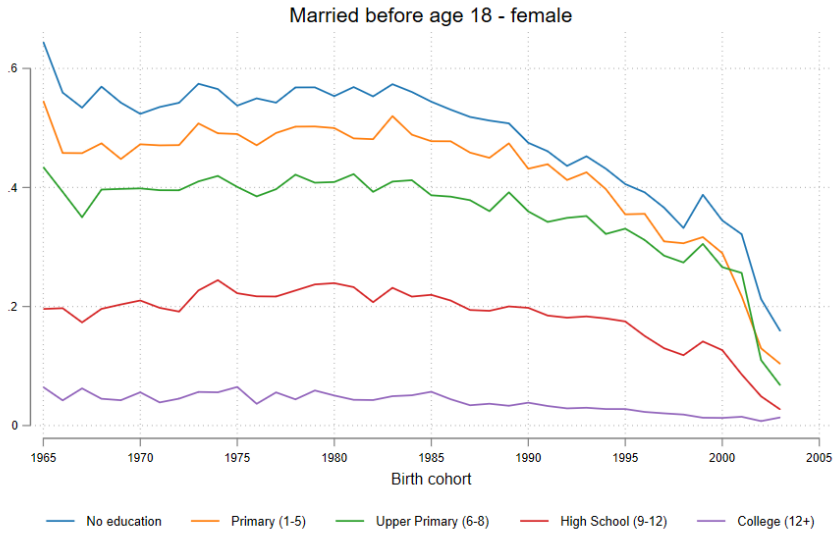

Figure 4: Child marriage

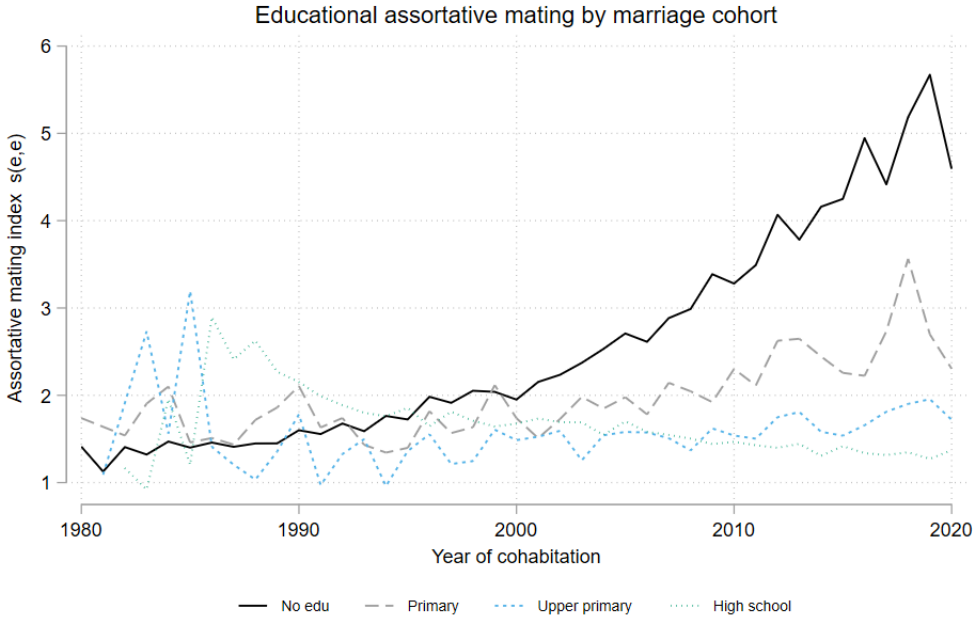

Figure 5: Assortative mating index (DHS 2019)

Table 1: Effect of School Construction on Women's Education Outcomes

	(1)	(2)	(3)
	Years of education	Primary completed	Secondary completed
Panel A: Continuous treatment			
Schools built	0.122***	0.019***	-0.001
	(0.021)	(0.002)	(0.002)
Mean	6.686	0.640	0.337
N	454,655	454,655	454,655
Panel B: Binary treatment			
Schools built > mean	0.433***	0.043***	0.007
	(0.107)	(0.008)	(0.011)
Mean	6.686	0.640	0.337
N	454,655	454,655	454,655

Notes: Each column reports estimates from regressions of the dependent variable on school construction exposure, with district and cohort fixed effects. Panel A uses cumulative schools constructed per 1,000 population aged 6–15 years. Panel B uses an indicator for whether exposure is above the cohort-specific mean. Standard errors (in parentheses) are clustered at the district level. Mean of cumulative school construction exposure in the estimation sample: 8.885.

Table 2: Effect of School Construction on Women's Marriage Outcomes

	(1)	(2)	(3)
	Age at marriage	Age gap	Not married by age 18
Panel A: Continuous treatment			
Schools built	0.036**	-0.025***	0.003**
	(0.015)	(0.009)	(0.002)
Mean	19.027	4.839	0.640
N	424,649	361,968	454,655
Panel B: Binary treatment			
Schools built > mean	0.282***	-0.102**	0.023**
	(0.080)	(0.042)	(0.010)
Mean	19.027	4.839	0.640
N	424,649	361,968	454,655

Notes: Each column reports estimates from regressions of the dependent variable on school construction exposure, with district and cohort fixed effects. In columns (1)–(2), the sample is restricted to ever-married women. Column (3) includes all women. Panel A uses cumulative schools constructed per 1,000 population aged 6–15 years. Panel B uses an indicator for whether exposure is above the cohort-specific mean. Standard errors (in parentheses) are clustered at the district level. Mean of cumulative school construction exposure in the estimation sample: 8.604.

Table 3: Effect of School Construction in Wife's District on Husband's Education Outcomes

	(1) Years of education	(2) Primary completed	(3) Secondary completed
Panel A: Continuous treatment			
Schools built	0.050	0.003	0.002
	(0.035)	(0.002)	(0.003)
Mean	8.065	0.746	0.416
N	63,815	63,815	63,815
Panel B: Binary treatment			
Schools built > mean	0.393	0.002	0.042***
	(0.250)	(0.014)	(0.013)
Mean	8.065	0.746	0.416
N	63,815	63,815	63,815

Notes: Each column reports estimates from regressions of the dependent variable on school construction exposure in the woman's district when young, with district and cohort fixed effects. Panel A uses cumulative schools constructed per 1,000 population aged 6–15 years. Panel B uses an indicator for whether exposure is above the cohort-specific mean. Standard errors (in parentheses) are clustered at the district level. Mean of cumulative school construction exposure in the estimation sample: 8.634.

Table 4: Effect of School Construction on Educational Assortative Mating Index

	(1) Assortative mating index
Schools built	0.030***
	(0.012)
Mean	2.104
N	4,646

Notes: District-by-marriage-cohort regression with district and cohort fixed effects. Cells weighted by number of couples. Mean of cumulative school construction exposure in estimation sample: 7.896.

A Appendix: District Panel Construction

Few studies in India undertake a long time series analysis at the district level due to changes in district boundaries over time. Some studies use only districts with unchanged boundaries which limits the sample, whereas other studies assume that district splits are subsets of the original parent district. However, in reality, new district boundaries can cut across multiple parent districts. This complicates the process of construction of a longer district level panel. One contribution of this study is the construction of a consistent district panel over a long time period 1961-2011. I use the 640 districts in the 2011 Census which also gives me a larger district level sample than other studies that only use unchanged districts.

If district boundary changes are not carefully accounted for, population data may have large measurement errors. To deal with this issue, I use the harmonized population counts across districts drawn from "Decadal Variation in Population" tables which account for district boundary changes over time which are provided by the Census of India. I then interpolate these population estimates across inter-censal years.

For data on outcomes, I use the 2019 Demographic and Health Survey (DHS). I map the shapefile for the 2019 DHS cluster geocodes to 2011 Census district administrative boundaries (available via SHRUG) using QGIS. My treatment variable is from the UDISE school database. I scraped geocodes for all 1.5 million schools and mapped it to DHS clusters using QGIS. I then map DHS households to DHS clusters which are now linked with Census district boundaries.

B Appendix: For Online Publication

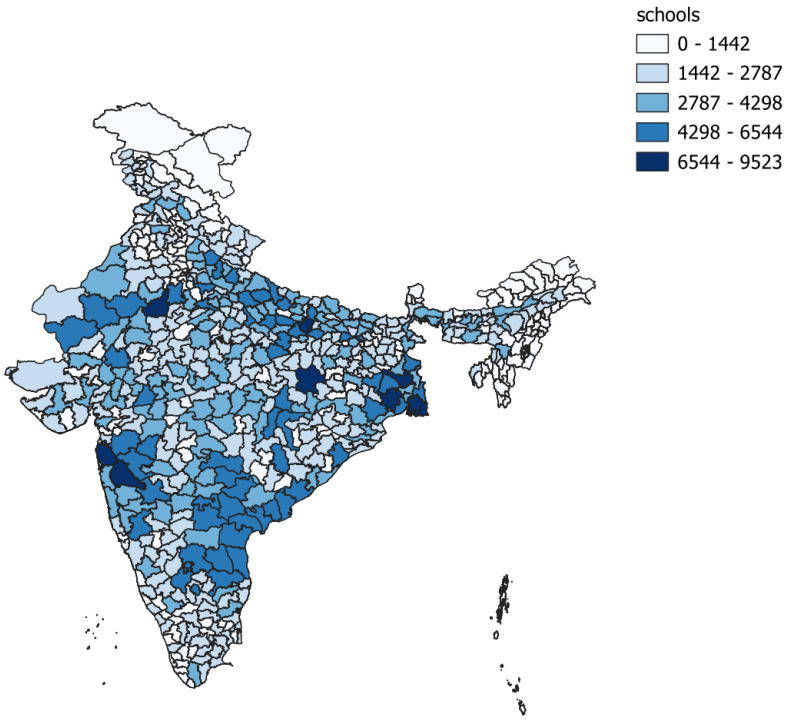

Figure 6: School construction across districts (UDISE 2018)

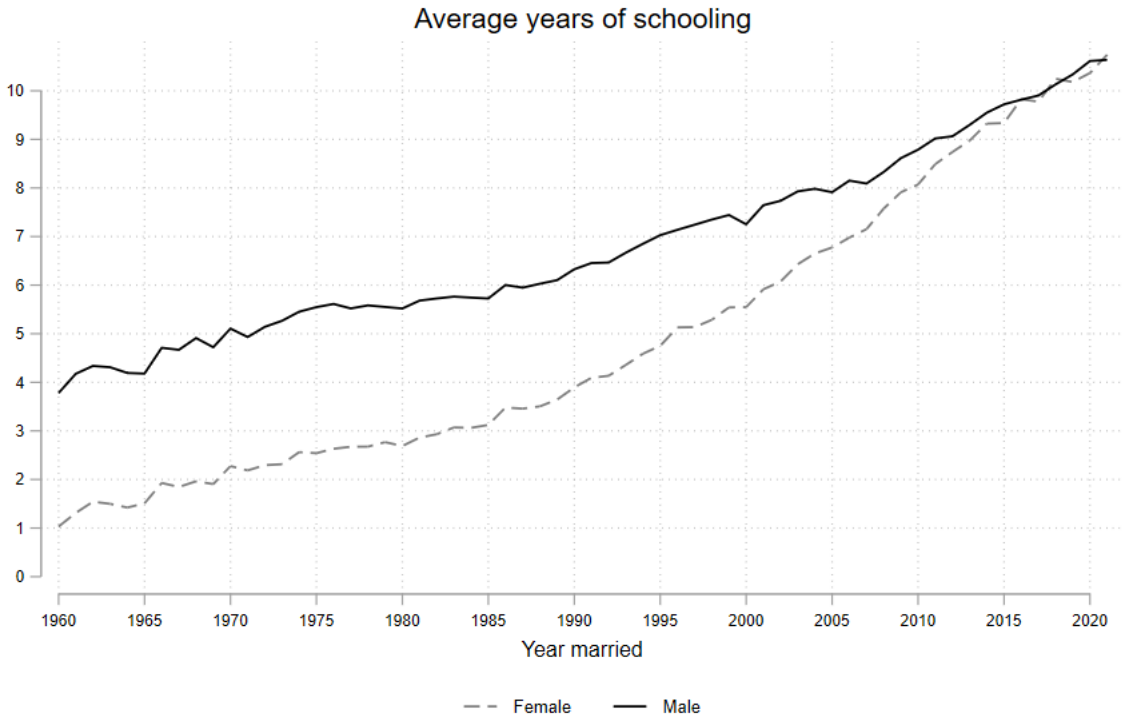

Figure 7: Average years of schooling among married couples

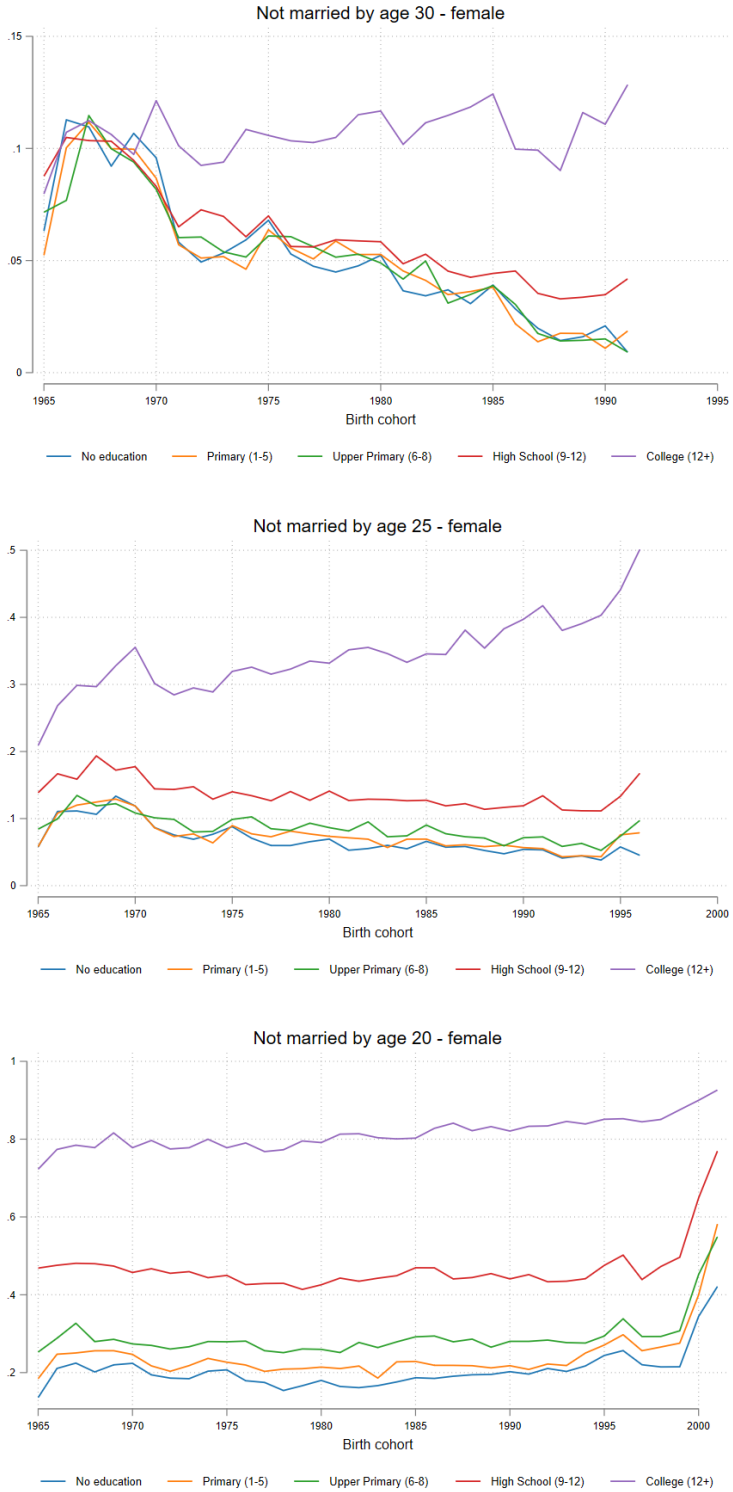

Figure 8: Not married by given age – women

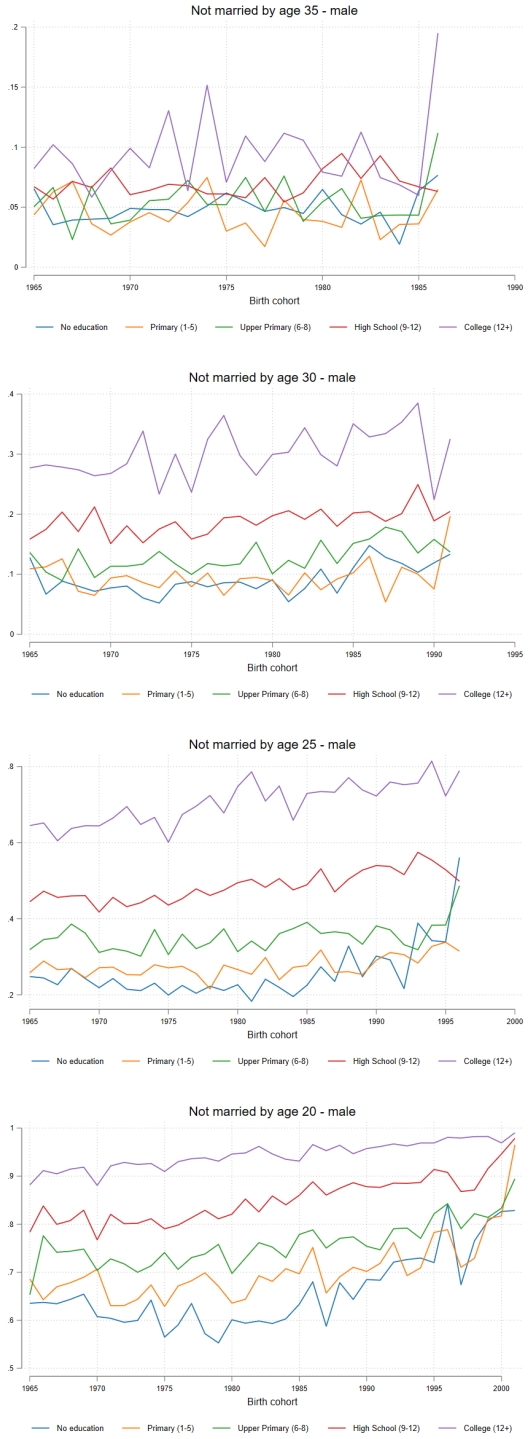

Figure 9: Not married by given age – men